# Student Statement

**Jaeho Jeon** · Ph.D. student, Department of Electrical Engineering and Computer Science, DGIST (Daegu, South Korea) · Privacy and Applied Cryptography Lab (PACL), advised by Prof. Young-Sik Kim · dgwogh@dgist.ac.kr

Presenting: **"Codeword Masking Can Be Harmful Under Replay Attacks on HQC"** (presenting author), accepted to Asiacrypt 2026

## Summary of research interests

My research focuses on **side-channel analysis** and the implementation security of cryptographic systems. I study how secret information leaks from devices through physical channels such as power consumption, and how countermeasures that look sound on paper can fail, or even backfire, when they meet a concrete attacker model. My recent work centers on **post-quantum cryptography (PQC)**, in particular code-based schemes: our Asiacrypt 2026 paper shows that codeword masking, a widely used countermeasure in HQC implementations, can actually be harmful under replay attacks, and analyzes the conditions under which it reduces rather than improves security.

My long-term goal is to become a recognized expert in side-channel analysis and to grow outward from it into computer architecture, networking, and coding theory, so that the security of a cryptographic system can be assessed end to end, from the algorithm down to the silicon.

## How I will benefit from participating in Asiacrypt

Asiacrypt 2026 will be my first time presenting at an IACR flagship conference. I am a first-year Ph.D. student, and I will present this paper on behalf of our team. Presenting it in person to the researchers who design, implement, and attack code-based schemes is the most direct way to test our claims, to learn which assumptions the community finds convincing, and to understand which follow-up questions matter most. Since HQC was selected by NIST for standardization, the question of which countermeasures are safe to deploy has become urgent, and I expect the discussions around our result to shape the next steps of my thesis.

Beyond my own talk, Asiacrypt is where the PQC, implementation-security, and side-channel communities meet. My lab is relatively small and geographically distant from most of these groups, so a few days of conversations at the conference are worth months of reading papers alone. What I learn at Asiacrypt will also be shared with the other members of my laboratory.

I would be grateful for the opportunity to present our work and to take part in the conference in person. Thank you for considering my application.

**Jaeho Jeon** · October 19, 2026

**DGIST · Privacy and Applied Cryptography Lab (PACL)**
Department of Electrical Engineering and Computer Science, Daegu Gyeongbuk Institute of Science and Technology, Daegu, South Korea

October 19, 2026

**To the Asiacrypt 2026 Student Stipend Committee**

**Re: Letter of support and justification of financial need for Jaeho Jeon**

I am writing to support the student stipend application of my Ph.D. student, **Jaeho Jeon**, who will present the paper *"Codeword Masking Can Be Harmful Under Replay Attacks on HQC"* at Asiacrypt 2026 on behalf of our team.

Jaeho joined my laboratory as a master's student in 2024 and began his Ph.D. in February 2026. He works on side-channel analysis and the implementation security of post-quantum cryptography. The Asiacrypt paper, which shows that codeword masking, a common countermeasure for HQC, can be harmful under replay attacks, is a joint work of our laboratory, and Jaeho will present it at the conference as the representative of the team. He is exactly the kind of early-career researcher who benefits most from presenting at, and meeting the community of, an IACR conference.

**Supplementary funding.** The laboratory will cover Jaeho's travel expenses for Asiacrypt 2026 in Hong Kong, including airfare and accommodation. However, we do not have funding available for the conference registration fee, and I would appreciate the committee's support for the registration cost.

I am confident that Jaeho will make full use of the opportunity, and that his participation will benefit both his development and the discussion of countermeasures for code-based PQC at the conference. Please do not hesitate to contact me if any further information is needed.

Sincerely,

**Young-Sik Kim**, Professor
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